

READING PASSAGE 2

Extremophiles and the Limits of Life

A Extremophiles and the Limits of Life has emerged as one of the most significant topics in contemporary academic discourse. The complexity and multifaceted nature of this subject have attracted the attention of researchers, policymakers, and practitioners across multiple disciplines, all seeking to understand its implications for the future.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D A substantial body of empirical evidence supports these theoretical frameworks. A landmark study conducted by researchers at leading international institutions analysed data spanning a thirty-year period and involving participants from more than forty countries. The findings revealed a consistent pattern of results that held across diverse cultural and economic contexts, suggesting that the identified mechanisms operate with considerable universality. The magnitude of the observed effects was particularly striking, with the key variables accounting for a substantial proportion of the total variance in outcomes.

E The practical application of research findings has proven more challenging than many initially anticipated. The translation of academic knowledge into actionable policy recommendations requires navigating a complex landscape of political constraints, economic interests, and social dynamics that academic models often simplify or overlook. Implementation failures in several high-profile initiatives have underscored the gap between theoretical understanding and practical capacity, prompting calls for more engaged and context-sensitive approaches to research.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G In conclusion, the study of extremophiles and the limits of life represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Questions 1–2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 2 The author notes that researchers have called for more _____ approaches to investigation.

Questions 3–4

Choose the correct letter, A, B, C or D.

- 3 According to the text, what role do structural factors play?
 - A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications
- 4 According to the text, what role do structural factors play?
 - A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 5 All critics agree that the research findings are universally applicable.

Question 6

Choose the correct letter, A, B, C or D.

- 6 The meta-analyses cited in the passage reveal that:
 - A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications

Question 7

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 7 The passage identifies power _____ as a key factor at the organisational level.

Questions 8–9

Choose the correct letter, A, B, C or D.

- 8** The meta-analyses cited in the passage reveal that:
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone
- 9** What does the passage identify as a significant challenge to current understanding?
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Question 10

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 10** The author notes that researchers have called for more _____ approaches to investigation.

Questions 11–13

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 11** Early researchers had access to the same technologies as modern scholars.
- 12** The landmark study discussed in the passage involved participants from multiple countries.
- 13** All critics agree that the research findings are universally applicable.